

RENO WARNER

AI PRODUCT & SYSTEMS ENGINEER

An innovative engineer solving complex, real-world problems by bridging high-performance physics-backed simulations with cutting-edge artificial intelligence. Specialized in developing serverless LLM applications, local edge machine learning inference engines in Rust/WASM, and high-reliability embedded systems.

EXPERIENCE

LLM Drone Analyzer

2026

Full-Stack AI Product | [Github](#) | [Live](#)

- Shipped a serverless web app where users configure drone powertrain parameters and receive instant AI-generated analysis backed by Monte Carlo simulation (1.5k+ runs).
- Wrote physics simulation entirely in Rust compiled to WASM
- Built LLM critique layer via OpenRouter: model receives distribution statistics and returns structured engineering recommendations; streamed to a React/Mantine UI with Chart.js.
- Full stack: React + TanStack Query frontend, Netlify serverless functions for LLM calls, Zod schema validation with unit-converting transforms.

Thrill Bot

2026

AI Recommendation Engine | [Github](#) | [Live](#)

- Built an LLM-powered recommender that translates natural language into structured database filters via a procedural Zod schema engine (createFilterSchema / applyFilter).
- Eliminated hallucination of invalid filter values through enum-preserving schema introspection fed directly to the tool-calling prompt.
- Integrated OpenRouter tool-calling via @openrouter/agent; frontend powered by TanStack Query + Netlify serverless functions.

Track Vision

2026

Edge Image ML in the Browser | [Github](#) | [Live](#)

- Deployed a MobileNetV2 image classifier running fully client-side: PyTorch → ONNX → Rust WASM (tract runtime). no server, no inference cost.
- Trained my own MobileNetV2 model to identify roller coaster manufactures. Achieved a 86% test accuracy score over a 20% random selection.
- Derived training sets from publically available sources.

EXPERIENCE

The Boring Company

Test Engineering Intern | Aug 2025 -

- **Automated Systems Verification:** Architected automated acceptance testing frameworks, bridging Python with Beckhoff TwinCAT to validate hydraulic manifolds and components against engineering truth models and schematics.
- **Reliability Software Development:** Developed a full-stack React web application for reliability tracking, enabling engineers to assign failure modes and mitigations, creating a closed-loop quality assurance system.
- **Failure Analysis:** Conducted root cause analysis and assigned failure modes for critical hydraulic and low-pressure fluid subsystems, including rock drill actuation, directly influencing design iterations for reliability.

SKILLS

- **Programming:** Matlab / Python, Rust, Git, Algorithms, Optimization, Performance, C, C++
- **Manufacturing:** Lean, CNC Mill / Lathe / 4th axis, CAM Fusion360, Laser / waterjet cutting
- **Analysis:** Finite Element Analysis (FEA), Computer-Aided Design(CAD), Constraints Engineering
- **Controls:** PID, Optimal Control, Model Predictive Control (MPC), Nonlinear Control
- **Embedded / Electronics:** ECAD, PCB Design, STM32, Arduino, FPGA, I2C, SPI, UART, LVDS

EXPERIENCE

Harbinger Small Satellite

Chief Engineer | Aug 2024 – May 2025

- **Systems Architecture:** Led a 16-person multidisciplinary team to deliver an ultra-low-cost (<\$20k) satellite. Designed the initial Concept of Operations (ConOps) and system architecture for a thermal optics payload.
- **Requirements & Compliance:** Established top-level Requirements Definitions and enforced strict adherence to Interface Control Documents (ICD) across power, payload, and comms subsystems.
- **Embedded Control Systems:** Developed the Attitude Control System (ACS) on STM32U5, implementing a gain-scheduled LQR quaternion-based controller utilizing Rust Embassy for efficient asynchronous execution without a traditional RTOS.

Utah Tech University

Manufacturing Engineering Intern | Aug 2022 – May 2024

- **Rapid Prototyping:** Collaborated with students and professors to rapidly transform high-level conceptual ideas into tangible, functional products.
- **System Reliability:** Developed dependable, quick-turn solutions for machine anomalies to maximize uptime and maintain low operations downtime.
- **Manufacturing Optimization:** Enhanced facility fabrication capabilities by designing custom tooling, fixtures, and specialized mechanical solutions.

EDUCATION

Florida Institute of Technology

Masters of Science in Space Systems |
Expected Q1 2027

- **Featured Courses:**
 - Space Craft Dynamics and Control
 - Launch / Space mission Operation
- **Focus:** Reliability and System Design

Utah Tech University

Bachelor of Science in Mechanical Engineering
| Emphasis Controls | May 2025

- **Featured Courses:** Autopilot, Modern Controls, Finite Element Analysis (FEA)
- **Certifications:**
 - ABET Accredited
 - NCEES FE Exam